

MODULE 01

BLAST & Sequence Alignment

eDNA Bioinformatics Learning Series · Yaw Baah

This module covers the foundational tool behind the entire eDNA pipeline: **BLAST** (Basic Local Alignment Search Tool). You will learn how sequence alignment works, what BLAST is doing under the hood, how to read its key output metrics, and why switching from web BLAST to a local installation is critical for large-scale and commercial environmental studies.

What you will learn

- ✓ What sequence alignment is and why it matters for eDNA
- ✓ How BLAST works step by step, in plain English
- ✓ How to read BLAST output: E-value, % identity, bit score, query coverage
- ✓ The difference between web BLAST and local BLAST+
- ✓ Where BLAST sits in your complete eDNA pipeline

1. The Problem BLAST Solves

When you collect an eDNA sample from a pond and sequence it, you get back thousands — sometimes millions — of short DNA sequences. Each one looks something like this:

ATCGATCGGAATCTACGATCGATCGATCGATCGATCGTACGATCGATCG...

There are no labels. No organism names. Just raw nucleotide letters. The fundamental question is: **which organism does this sequence come from?** BLAST answers this by comparing your unknown sequence against a database of hundreds of millions of sequences from known organisms.

Analogy — The Forensic Library

Imagine finding a page torn from an unknown book. To identify it, you visit a library and compare that page against every book on the shelves. When you find the closest match — even if not word-for-word identical — you can say: *this page probably came from that book*. **BLAST is the librarian performing that search for DNA sequences.** The library is the NCBI nucleotide database — currently over 500 billion bases from millions of known organisms worldwide.

2. What is Sequence Alignment?

Alignment means lining up two sequences so matching bases sit above each other. Gaps (-) are inserted where one sequence has an insertion or deletion the other lacks.

Query : ATCG-ATCGATCG

|||| |||

Subject: ATCGAATCGATCG

12 of 13 positions match — 92.3 % identity. The dash is a gap (one base insertion in subject).

Type	What it does	When used
Global	Aligns end-to-end across full length	Closely related sequences of similar length
Local	Finds best-matching region — even a sub-segment	BLAST uses this — your eDNA read may only match part of a long database entry

3. How BLAST Works — Three Stages

A naive approach would check every sequence in the database — impossible at 500 billion bases. BLAST uses a three-stage shortcut to be fast without sacrificing accuracy:

1. Seeding

BLAST breaks your query into short overlapping words called **k-mers** (default: 11 bases). It scans the database for exact matches to these words using an index — like looking up a term in a book's index rather than reading every page.

2. Extension	Each seed match is extended in both directions, one base at a time. Extension continues while the alignment score improves. When the score drops below a cutoff, it stops — preventing wasted time on poor matches.
3. Scoring and Reporting	Each extended match (called an HSP — High-Scoring Pair) receives a score based on: identical bases (+), mismatches (-), gap penalties. BLAST calculates statistical confidence and reports all alignments above your threshold, ranked by score.

4. Reading BLAST Output — The Four Key Numbers

Every BLAST hit returns four numbers. Your pipeline stores all of them in **blast_hit_table.csv**. Knowing what they mean is essential to interpreting biodiversity results correctly.

% Identity (**identity_pct**)

The percentage of aligned bases that are *identical* between your query and the database match.

Value	Interpretation
97 – 100 %	Species-level assignment reliable
90 – 96 %	Genus-level reliable; species uncertain
80 – 89 %	Family-level only; treat with caution
< 80 %	Too divergent for confident classification

Pipeline note:

Your pipeline uses this to set the **confidence_flag** column (high / medium / low) and drives the LCA downgrading logic in `resolve_taxonomy()`.

E-value (**eval**)

The number of times you would expect to see a match this good *by chance* in a database of this size. Lower = more significant.

Value	Interpretation
1e-100	Near-certain match — extremely significant
1e-10	Significant — your pipeline default threshold
1e-5	Weak but possibly real — use with caution
> 0.01	Likely random noise — discard

Pipeline note:

Think of E-value like a p-value in statistics. Your pipeline filters out anything above 1e-10 by default via the **--eval** flag.

Bit Score (**bit_score**)

A normalised score that accounts for the scoring matrix used. Higher is always better. Unlike raw scores, bit scores are comparable across different database sizes.

Value	Interpretation
> 200	Strong alignment

50 – 200	Moderate alignment
< 50	Weak — likely unreliable

Pipeline note:

Your pipeline uses bit score to select the **best HSP** when a single alignment has multiple non-contiguous matching segments.

Query Coverage % (query_coverage_pct)

The percentage of your query sequence covered by the alignment. High identity on a tiny fragment is far less meaningful than high identity across the full sequence.

Value	Interpretation
> 90 %	Good — full-length match
50 – 90 %	Partial match — interpret carefully
< 50 %	Fragment only — low confidence

Pipeline note:

Your pipeline clamps this to 100 % maximum. Subject insertions can push the raw value above 100, which is mathematically meaningless.

5. Web BLAST vs Local BLAST+

Your original pipeline used NCBI's web BLAST service. Phase 1 of the commercial build replaces this with a local BLAST+ installation. Here is exactly why that matters:

Property	Web BLAST (original)	Local BLAST+ (Phase 1)
Speed per sequence	30 – 120 seconds	0.01 – 1 second
10,000 sequences	~9 hours	5 – 15 minutes
Rate limit	3 requests/sec (NCBI enforced)	No limit — uses your CPU
Reproducibility	Database updates silently	Database version locked and logged
Regulatory compliance	Cannot cite 'NCBI as of today'	Version stamped in every output CSV
Internet required	Yes — fails offline	No — fully local
Cost	Free but limited	Free, unlimited, scales with hardware

Why reproducibility is non-negotiable for commercial work

If you run your pipeline today and again in six months, NCBI will have added new sequences. You could get different species assignments for the same input — not because the biology changed, but because the database did. A client comparing two reports would see inconsistencies that look like errors. **Local BLAST+ with a versioned database eliminates this entirely.** The database version is stamped into every output file — the same way a lab records the reagent batch number for regulatory traceability.

What changes in the code

The science is *identical* — same algorithm, same scoring, same XML output. Only the transport layer changes: instead of an HTTP request to NCBI, Python calls the local **blastn** binary via subprocess:

```
# BEFORE – web BLAST (network call to NCBI)
result = NCBIWWW.qblast('blastn', 'nt', sequence)

# AFTER – local BLAST+ (subprocess call to binary)
blastn -query seq.fasta -db /data/nt -outfmt 5 -num_threads 8
```

What -num_threads 8 does

BLAST is *embarrassingly parallel* — each query sequence is independent of the others. The **-num_threads** flag tells BLAST to process multiple sequences simultaneously across CPU cores. On a modern 8-core laptop this gives a 6-8x speed boost. On a 32-core cloud server you can classify tens of thousands of sequences in minutes.

6. Where BLAST Sits in Your Pipeline

Full data flow from a pond water sample to a classified species report, showing exactly where BLAST fits:

#	Stage	What happens
1	Sample collection	Pond water filtered through a membrane. DNA extracted and amplified by PCR using marker-specific primers (16S rRNA for bacteria, 18S rRNA for eukaryotes, ITS for fungi, COI for invertebrates).
2	Sequencing	Illumina or Oxford Nanopore reads the amplified DNA. Output: millions of short sequences in FASTQ format, each with a quality score per base.
3	Quality filtering	Low-quality reads, chimeric sequences, and primer artefacts removed. Output: clean FASTA file ready for identification.
4	BLAST <-- YOU ARE HERE	Each clean sequence is aligned against the reference database. Output: XML file with top hits, scores, and E-values per query sequence.
5	Parsing + LCA	src/parser.py extracts species names from BLAST XML. resolve_taxonomy() applies Lowest Common Ancestor logic to determine the most confident taxonomic level.
6	Diversity metrics	Shannon H', Pielou J', and species richness calculated from resolved taxonomy. Output: biodiversity_summary.csv
7	PDF report	Formatted report generated with species table, confidence flags, protected species alerts, and diversity charts. (Phase 2)

7. Glossary

% identity

Percentage of aligned bases that are identical between query and subject.

16S rRNA

Bacterial ribosomal gene — universal prokaryote barcode.

18S rRNA

Eukaryotic ribosomal gene — used for algae, protists, micro-eukaryotes.

Alignment

Arranging two or more sequences to identify regions of similarity.

ASV

Amplicon Sequence Variant — exact sequence produced by denoising; higher resolution than OTUs.

Bit score

Normalised alignment score — higher is better, comparable across database sizes.

BLAST

Basic Local Alignment Search Tool — finds similar sequences in a database.

blastn

BLAST program for nucleotide vs nucleotide searches (DNA query, DNA database).

COI

Cytochrome c Oxidase subunit I — mitochondrial gene used as a barcode for animals.

E-value

Expect value — expected number of random hits with that score. Lower = more significant.

eDNA

Environmental DNA — genetic material shed into surroundings (water, soil, air).

HSP

High-Scoring Pair — a local alignment found by BLAST. One alignment can have multiple HSPs.

ITS

Internal Transcribed Spacer — ribosomal DNA region used as a barcode, especially for fungi.

k-mer

Short sequence of k bases used as a seed in BLAST's initial scanning step.

LCA

Lowest Common Ancestor — assigns taxonomy at the deepest level all top hits agree on.

Metabarcoding

Sequencing and identifying DNA from mixed environmental samples.

OTU

Operational Taxonomic Unit — cluster of sequences at 97 % identity, used as a species proxy.

Query coverage

Percentage of your query sequence spanned by the BLAST alignment.

Key Takeaways

■ BLAST is a seed-and-extend algorithm, not brute force

It finds short exact k-mer matches first, then extends them — making it thousands of times faster than aligning against every database sequence naively.

■ E-value is your primary significance filter

It tells you the statistical probability of a random hit. Your pipeline defaults to 1e-10. Anything above 0.01 is likely noise.

■ 97 % identity is the species-level threshold

Below 97 %, the sequence is too divergent to reliably name a species for most eDNA barcoding markers (16S, 18S, ITS, COI). Your pipeline's LCA logic respects this boundary automatically.

■ Web BLAST is not reproducible — local BLAST+ is

Database updates silently on NCBI. For commercial and regulatory work, you need a version-locked local database with the version number stamped in your outputs.

■ Local BLAST+ is the same science, faster delivery

Same algorithm, same scoring matrix, same XML output format — just runs on your hardware instead of NCBI's servers. Your pipeline wraps this transparently behind `--local`.